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Software Requirements Specification

for

RouteX: A Truck Profitability and Delivery Operations Management System

Version 1.0

Prepared for:
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Revision History

Name	Date	Reason For Changes	Version
RouteX	09/26/2026	Initial SRS Draft	1.0

1. Introduction

1.1. Document Conventions

This Software Requirements Specification (SRS) uses consistent headings and numbering to organize the requirements of the RouteX system. The following conventions are used throughout the document:

Convention	Description
RouteX	Refers to the proposed **RouteX: A Truck Profitability and Delivery Operations Management System** for Redeemed Route Logistics Inc.
SRS	Refers to the Software Requirements Specification document.
POD	Refers to Proof of Delivery documents associated with delivery transactions.
RBAC	Refers to Role-Based Access Control, which restricts system access based on the assigned role and permissions of an authorized user.
User	Refers to an authorized person who is permitted to access and use the RouteX system.
System	Refers to the proposed RouteX web application.
Shall	Indicates a required system behavior or requirement.
Should	Indicates a recommended system behavior or requirement.
May	Indicates an optional system behavior or requirement.
REQ	Identifies a functional requirement in the SRS.
NFR	Identifies a nonfunctional requirement in the SRS.
TBD	Indicates information that has not yet been finalized.

Requirement statements are written in clear and testable terms to describe what the RouteX system shall provide.

1.2. Purpose

The purpose of this Software Requirements Specification is to define the requirements and expected functions of **RouteX: A Truck Profitability and Delivery Operations Management System** for **Redeemed Route Logistics Inc.**

RouteX is a proposed web-based and partially automated system intended for the internal use of authorized personnel of Redeemed Route Logistics Inc. The system is intended to improve the management of the company's trucking delivery operations and related financial information through centralized operational and financial records.

The proposed system will provide authorized personnel with functions for recording and managing delivery transactions, monitoring delivery information, recording revenue and expenses, calculating delivery profitability, searching and retrieving records, generating operational and financial reports, managing Proof of Delivery (POD), and maintaining secure system access.

RouteX is intended to improve the company's existing manual processes, which primarily rely on **Microsoft Excel, basic bookkeeping, and other manual methods** for managing operational and financial information.

This SRS serves as a reference for the development, testing, evaluation, and acceptance of the proposed RouteX system.

1.3. Project Scope

1.3.1. Scope

RouteX is a proposed **web-based and partially automated Truck Profitability and Delivery Operations Management System** for Redeemed Route Logistics Inc. The system will provide a centralized platform for managing the company's delivery and financial records.

The scope of RouteX covers the following areas:

1. **Delivery Management**
    - Create, update, and manage delivery records.
    - Record delivery ID, truck plate number, assigned driver, client information, delivery route, delivery date, and delivery status.
    - Record delivery revenue and delivery-related expenses.
    - Upload and manage Proof of Delivery (POD) documents.
    - Automatically calculate delivery margin based on recorded revenue and expenses.
2. **Transactions Management**
    - Record and manage company expenses that are not directly associated with a specific delivery.
    - Maintain records of business-related expenses such as insurance, vehicle registration, office supplies, utilities, equipment purchases, and other company expenses.
3. **Search and Record Retrieval**
    - Search, filter, and retrieve delivery and transaction records.
    - Search using delivery ID, truck plate number, driver, client, and date.
    - Retrieve historical operational and transaction records.
4. **Reports and Profitability Analysis**
    - Generate delivery activity reports.
    - Generate revenue and expense reports.
    - Generate profitability reports.
    - Provide printable and exportable reports for management and documentation purposes.
5. **Security and Administration**
    - Provide user authentication.
    - Implement role-based access control.
    - Manage user accounts and permissions.
    - Maintain activity logs and audit trails.

RouteX is **partially automated**. Authorized users will manually enter and update the required information, while the system will assist with repetitive activities such as organizing records, calculating delivery margins, searching and retrieving information, and generating reports.

1.3.2. Limitations

The proposed RouteX system has the following limitations:

- The system will not provide automatic real-time GPS tracking of trucks.
- The system will not automatically obtain truck location information through GPS or tracking devices.
- The system will not include integration with GPS or other vehicle tracking devices.
- The system is limited to the delivery management, transaction management, search and record retrieval, reporting and profitability analysis, and security and administration functions included in the agreed project scope.
- The accuracy of records and generated reports depends on the accuracy and completeness of information entered and updated by authorized users.
- Features outside the agreed project scope are not included in the proposed system.

1.4. Product Perspective

Redeemed Route Logistics Inc. currently does not have a formal or integrated information system for managing its delivery operations and related financial information. Its current processes primarily involve **Microsoft Excel, basic bookkeeping, and manual communication and documentation methods**.

RouteX is intended to provide a centralized web-based system for managing delivery operations and truck profitability. Authorized personnel can enter and manage delivery and expense information, which the system processes for delivery monitoring, profitability calculation, record retrieval, and report generation.

RouteX is **partially automated**. Users remain responsible for entering and updating information, while the system assists in organizing, processing, calculating, retrieving, and reporting the recorded data.

The proposed system uses:

- **Frontend:** Next.js, React, TypeScript, Tailwind CSS
- **Backend:** Next.js Server Actions + API Routes
- **Database:** PostgreSQL through Supabase
- **Authentication:** NextAuth.js / Auth.js
- **File Storage:** Supabase Storage
- **Deployment:** Vercel

1.5. Product Functions

RouteX will provide the following main functions:

1. **Delivery Management** – Manage delivery records, delivery status, revenue, delivery expenses, and Proof of Delivery (POD) documents.
2. **Transactions Management** – Record and manage company expenses not associated with a specific delivery.
3. **Search and Record Retrieval** – Search, filter, and retrieve delivery, transaction, and related records.
4. **Reports and Profitability Analysis** – Generate operational, financial, and profitability reports.
5. **Security and Administration** – Manage authentication, user access, permissions, and activity logs.

1.6. Design and Implementation Constraints

1.6.1. Technical Constraints

- The system shall be implemented as a web-based application accessible through a modern web browser.
- The frontend shall use Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, and Lucide React.
- The backend shall use Next.js Server Actions and API Routes.
- The database shall use PostgreSQL through Supabase.
- The system shall use Supabase Storage for uploaded files and supporting documents.
- Authentication shall use NextAuth.js (Auth.js).
- The system shall be deployed using Vercel.
- The system requires a stable internet connection.
- The minimum hardware requirement shall include a dual-core 2.0 GHz processor or higher and 8 GB RAM, with at least 1 GB of available storage and a display resolution of 1366 × 768 or higher.
- The system shall be accessed using a modern web browser.

1.6.2. Operational and Scope Constraints

- Access to the system shall be restricted to authorized users.
- Users shall be responsible for entering and updating the required delivery and transaction information.
- Delivery information shall depend on staff-entered updates.
- The system shall not provide automatic real-time GPS tracking of trucks.
- The system shall not include automatic GPS or tracking-device integration.
- The system shall operate only within the agreed project scope and requirements.
- Features outside the agreed project scope shall not be included in the proposed system.

- The accuracy of records and generated reports shall depend on the accuracy and completeness of information entered by authorized users.

1.7. User Documentation

The following documentation and orientation materials will be provided to support the proper use and operation of the RouteX system:

1. **User Manual** – Provides instructions on how authorized users can access and use the system features, including delivery management, transaction management, search and record retrieval, report generation, and other functions based on the user's assigned role.
2. **User Orientation/Training Materials** – Provides guidance and supporting materials for training authorized personnel on the proper use of RouteX during system turnover.
3. **Technical Documentation** – Provides technical information about the developed system to support system understanding and maintenance.

The documentation will be provided in digital format and will be based on the final implemented version of RouteX.

1.8. Assumptions and Dependencies

1.8.1. Assumptions

The following assumptions apply to the proposed RouteX system:

1. The client will provide the necessary information and resources required for system development, implementation, testing, and turnover.
2. Authorized users will provide accurate and complete information when recording and updating delivery and transaction records.
3. Only authorized personnel will be provided with system credentials.
4. Users will properly safeguard their account credentials and will not share them with unauthorized persons.
5. Users will have access to a modern web browser and a stable internet connection when using RouteX.
6. Authorized users will be properly oriented or trained before using the system.
7. The agreed project requirements and scope will serve as the basis for system development and implementation.

1.8.2. Dependencies

The proposed RouteX system depends on the following:

1. **Next.js, React, TypeScript, and Tailwind CSS** for frontend development.

2. **Next.js Server Actions and API Routes** for backend processing.
3. **PostgreSQL through Supabase** for database storage.
4. **Supabase Storage** for uploaded files and supporting documents.
5. **NextAuth.js / Auth.js** for user authentication.
6. **Vercel** for system deployment.
7. A stable internet connection for accessing the online web application.
8. Client cooperation in providing requirements, feedback, testing participation, and necessary resources.

2. Data Requirements

2.1. Data Validation Rules

- **Delivery ID, truck plate number, driver name, and client name** shall be required when creating a delivery record.
- **Delivery dates** shall use a valid date format.
- **Revenue and expense amounts** shall be numeric and non-negative.
- **Delivery margin** shall be automatically calculated from revenue and delivery expenses and shall not be directly editable.
- **Delivery status** shall be limited to predefined values, such as Pending, In Transit, Delivered, and Cancelled.
- **Proof of Delivery (POD)** uploads shall use supported file types and size limits and shall be linked to an existing delivery record.
- **User accounts** shall use unique usernames or email addresses, and passwords shall follow the required security rules.

2.2. Data Security and Privacy Requirements

- The system shall implement **authentication and role-based access control** so that only authorized users can access and manage records.
- Passwords shall be stored using a **one-way cryptographic hash** and shall not be stored or displayed in plain text.
- Data transmitted between the client browser and server shall be **encrypted in transit using HTTPS/TLS**.
- The system shall comply with the **Data Privacy Act of 2012 (RA 10173)** and applicable data protection requirements.
- User activities, including record creation, modification, deletion, and login events, shall be recorded through **audit trails and activity logs**.
- Access to client data shall be limited to purposes related to the **development, maintenance, and operation** of RouteX and shall be kept confidential.

2.3. Data Retention and Backup Requirements

- Delivery, transaction/expense, user, and POD records shall be retained to support operational use, historical record retrieval, and reporting.
- RouteX data shall be backed up to reduce the risk of data loss due to system failure or other incidents.

- The specific **data retention period and backup schedule are TBD** and shall be finalized based on the client's requirements and system implementation considerations.
- Upon termination of the project or agreement, client data shall be handled according to the agreed terms between the client and provider.

3. System Features

3.1. Functional Requirements

3.1.1. System Feature #1

3.1.2. System Feature #2

3.2. Nonfunctional Requirements

3.2.1. Performance Requirements

3.2.2. Safety Requirements

3.2.3. Security Requirements

3.2.4. Software Quality Attributes

3.2.5. Business Rules

4. System Model and Diagrams

4.1. Use Case Diagrams

4.2. Entity Relationship Diagrams

4.3. Activity Diagrams

5. Other Requirements

Appendix A: Glossary

Appendix B: To Be Determined List

Appendix C: References

Requirements Validation / Consistency Review